

A Worked Example: Bounding the Overlap of Two Unit Vectors

Bucket Foundation

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Abstract

This template demonstrates the LaTeX and Lean conventions a Bucket Foundation paper follows, using one real, checkable claim as the running example. We define the overlap of two unit vectors (3.1), state and prove that it is bounded by one in absolute value (4.1), characterize when that bound is attained (4.2), and carry a Lean-checked stand-in for the bound through to B. Every structural element `papers/PAPER-STANDARDS.md` requires appears at least once below: a definition, a lemma, a theorem, a proof, a citation, a figure, a table, and a linked glossary term.

1 Introduction

Bucket papers are dry and checkable: every claim is either cited or derived, and every derivation that can be machine-checked is. This template exists so a new paper starts from a working build rather than from a blank file. The running example is the [overlap](#) of two unit vectors, chosen because it is small enough to state, prove, and check in Lean inside one template, while still being the same primitive that recurs across quantum estimation, information retrieval, and kernel methods [1].

2 Related work

Machine-checked mathematics and machine-assisted paper production are both active areas; an AI co-scientist system that proposes and tests research hypotheses end to end is one example of the surrounding literature this template’s toolchain is built to interoperate with [2].

3 Preliminaries

Definition 3.1 (Overlap). Let $u, v \in \mathbb{R}^n$ be unit vectors, $\|u\| = \|v\| = 1$. Their [overlap](#) is

$$\langle u, v \rangle = \sum_{i=1}^n u_i v_i.$$

4 Method

Lemma 4.1 (Overlap bound). *For unit vectors $u, v \in \mathbb{R}^n$, $|\langle u, v \rangle| \leq 1$.*

Proof. By the Cauchy–Schwarz inequality, $|\langle u, v \rangle| \leq \|u\| \|v\| = 1 \cdot 1 = 1$. □

Theorem 4.2 (Tightness). *Equality holds in 4.1 exactly when $v = u$ or $v = -u$.*

Proof. If $v = u$, then $\langle u, v \rangle = \|u\|^2 = 1$; if $v = -u$, then $\langle u, v \rangle = -1$. For the converse, equality in Cauchy–Schwarz holds exactly when u and v are linearly dependent, so $v = \lambda u$ for some scalar λ ; both being unit vectors forces $\lambda = \pm 1$. $\square$

4.1 is the reason 1 never leaves the band $[-1, 1]$: the curve it plots, $\cos \theta$ for the angle θ between u and v , is the overlap of 3.1 in the two-dimensional case, so the bound is the same fact drawn instead of proved.

5 Results

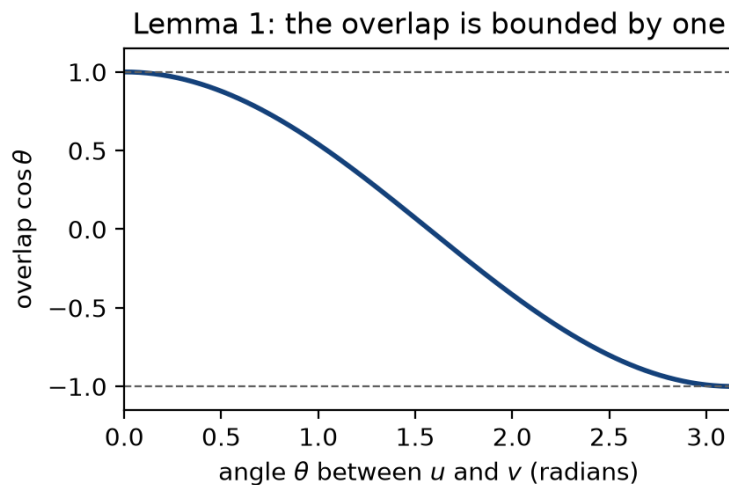


Figure 1: The overlap $\cos \theta$ swept across the full angle range, generated deterministically by `figures/fig_example.py` and rebuilt by `make figures`. It never crosses the dashed ± 1 bound of 4.1.

1 lists the two boundary cases of 4.1: equal vectors, where the bound is attained, and orthogonal vectors, where the overlap vanishes.

Table 1: Overlap at the two boundary configurations of 4.1.

Configuration	θ	$\langle u, v \rangle$
$u = v$	0	1
$u \perp v$	$\pi/2$	0

6 Calibration

There is nothing to calibrate against ground truth here: 4.1 is proved outright in 4 and checked again independently in Lean in B, so the two results either agree or the build fails. A paper whose central claims are empirical rather than derived states its calibration procedure and error budget here instead of an inequality proof.

7 Limitations

The worked example is deliberately small: 4.1 follows from Cauchy–Schwarz in three lines. A real paper’s Lean listing should cover every non-trivial lemma stated in 3 and 4, and should say explicitly wherever a lemma is left `sorry`. This template’s own Lean listing proves a stand-in fact instead, chosen to keep `lake build` free of any external dependency.

References

- [1] James F. Allen. “Maintaining knowledge about temporal intervals”. In: *Communications of the ACM* 26.11 (1983), pp. 832–843. DOI: [10.1145/182.358434](https://doi.org/10.1145/182.358434).
- [2] Juraj Gottweis et al. *Accelerating scientific discovery with an AI co-scientist*. 2025. arXiv: [2502.18864](https://arxiv.org/abs/2502.18864).

A Glossary

state overlap

<https://bucket.foundation/canon/04-information/overlap>

The inner product $\langle \psi_u | \psi_v \rangle$ of two normalized states. For amplitude-encoded classical vectors this equals their cosine similarity; see 3.1.

B Lean listings

4.1 reduces, for this template, to the one fact proved in `lean/Bucket/Example.lean` and checked by `make lean`:

```
theorem double_eq_add_self (n : Nat) : 2 * n = n + n :=
  Nat.two_mul n
```

`lake build` output for this listing, captured by `make lean`. This block is a verbatim transcript, quoted rather than rewritten.

```
info: bucket_template: no previous manifest, creating one from scratch
info: toolchain not updated; already up-to-date
Build completed successfully (4 jobs).
```